United International University (UIU)
Dept. of Electrical and Electronic Engineering (EEE)
Course No. EEE 312
Course Title: Digital Signal Processing Lab

Lab Sheet 7
Digital Filter Design Using MATLAB Toolbox

Prerequisite

Before starting this section, students should know..

1. A fundamental knowledge on MATLAB
2. A theoretical knowledge on digital filter.

Outcomes

After finishing this lab students should be able to.....

1. design digital filter using MATLAB toolbox.
2. understand different characteristics of Digital Filter.

Outlines of this Lab

0. Introduction
1. Lab Session 7.1: FIR and IIR filter design Using fvtool toolbox
2. Lab Session 7.2: Filter Design Using fdatool
3. Lab Session 7.3: In Lab Evaluation
4. Home work

Introduction:

Methods of the design of FIR and IIR filters.

1. Using fvtool toolbox
2. Using fdatool

Lab Session 1.1: FIR and IIR filter design Using fvtool toolbox

Used function : designfilt

Syntax

```
d = designfilt(resp,Name,Value)
fvtool (d)
```


First OBE version, prepared by: B.K.M. Mizanur Rahman, Assistant Professor, Department of EEE, UIU

About the input arguments:

d = designfilt(resp, Name, Value)

Filter response and type: any one of the following
'lowpassfir'
'lowpassiir'
'highpassfir'
'highpassiir'
'bandpassfir'
'bandpassiir'
'bandstopfir'
'bandstopiir'
'differentiatorfir'
'hilbertfir'
'arbmagfir'

Value Pair Arguments
Specify optional comma-separated pairs of Name, Value arguments. Name is the argument name and Value is the corresponding value. Name must appear inside single quotes (' '). You can specify several name and value pair arguments in any order as
Name1,Value1,...,NameN, ValueN.

Example: 'FilterOrder',20,'CutoffFrequency',0.4 suffices to specify a lowpass FIR filter.

Some Terminology:

Name	Meaning
'FilterOrder'	Some design methods let you specify the order. Others produce minimum-order designs. That is, they generate the smallest filters that satisfy the specified constraints.
'DesignMethod'	It is the algorithm used to design the filter. Examples include constrained least squares ('cls') and windowing ('window'). For some specification sets, there are multiple design methods available to choose from. In other cases, you can use only one method to meet the desired specifications.
'SampleRate'	It is the frequency at which the filter operates. designfilt has a default sample rate of 2 Hz. Using this value is equivalent to working with normalized frequencies.
Design options	They are parameters specific to a given design method. Examples include 'Window' for the windowing method and optimization 'Weights' for arbitrary-magnitude equiripple designs. (See the complete list under Name-Value Pair Arguments.) designfilt provides default values for design options left unspecified.

Example 7.1: Design a lowpass FIR with the following specifications:

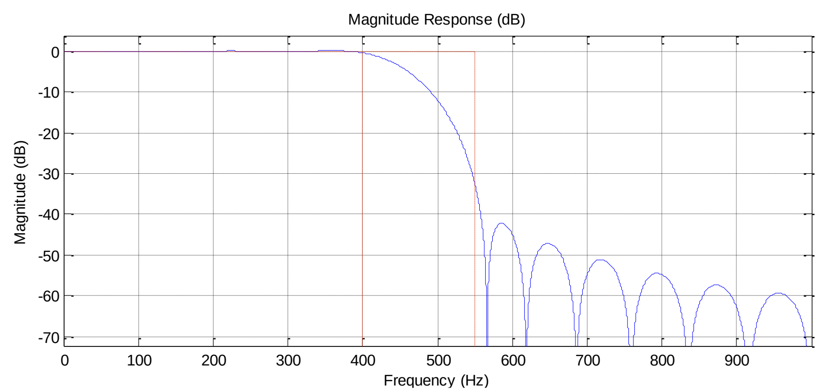
FrequencyResponse: 'lowpass'
ImpulseResponse: 'fir'
SampleRate: 2000
FilterOrder: 25
PassbandFrequency: 400
StopbandFrequency: 550

DesignMethod: 'ls'

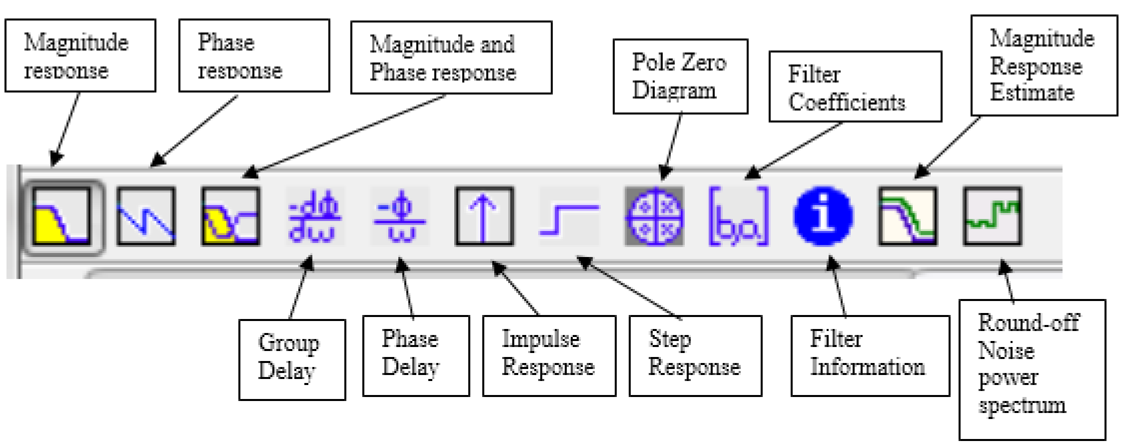
Solution: MATLAB Code:

```
d = designfilt('lowpassfir', ...          % Response type
       'FilterOrder',25, ...              % Filter order
       'PassbandFrequency',400, ...       % Frequency constraints
       'StopbandFrequency',550, ...
       'DesignMethod','ls', ...           % Design method
       'PassbandWeight',1, ...            % Design method options
       'StopbandWeight',2, ...
       'SampleRate',2000);                 % Sample rate
   fvtool(d)
```

Output:



Note: In the figure window observe the following portion. You can click any menu for corresponding output:



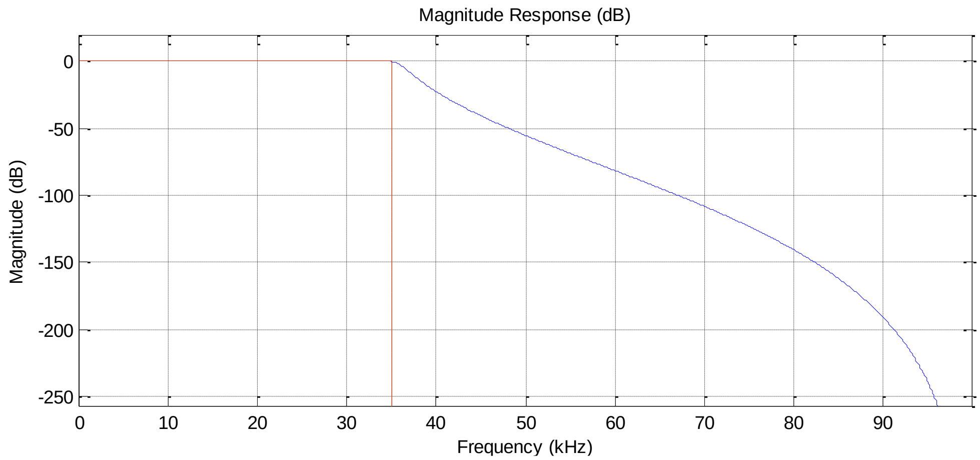
Example 7.2: Design a lowpass IIR with the following specifications:

SampleRate: 200000
FilterOrder: 8
PassbandFrequency: 35000
PassbandRipple: 0.2000
DesignMethod: 'cheby1'

MATLAB Code:

```
lpFilt = designfilt('lowpassiir','FilterOrder',8, ...
         'PassbandFrequency',35e3,'PassbandRipple',0.2, ...
         'SampleRate',200e3)
fvtool(lpFilt)
```

Output:



Example 7.3: Design a highpass FIR with the following specifications:

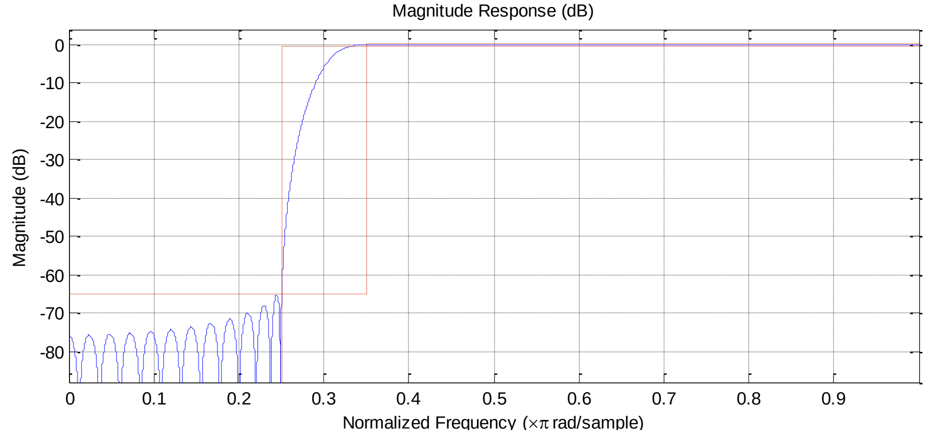
SampleRate: 2
StopbandFrequency: 0.2500
PassbandFrequency: 0.3500
StopbandAttenuation: 65
PassbandRipple: 0.5000
DesignMethod: 'kaiserwin'

Solution:

MATLAB Code:

```
hpFilt = designfilt('highpassfir','StopbandFrequency',0.25, ...
         'PassbandFrequency',0.35,'PassbandRipple',0.5, ...
         'StopbandAttenuation',65,'DesignMethod','kaiserwin')
fvtool(hpFilt)
```

Output:



Example 7.4: Design a highpass IIR with the following specifications:

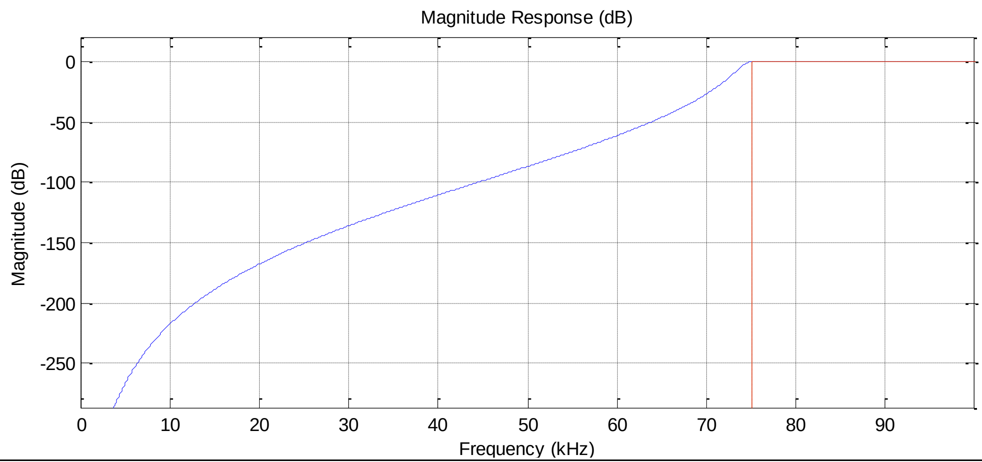
SampleRate: 200000
FilterOrder: 8
PassbandFrequency: 75000
PassbandRipple: 0.2000
DesignMethod: 'cheby1'

Solution: MATLAB Code:

```
hpFilt = designfilt('highpassiir','FilterOrder',8, ...

'PassbandFrequency',75e3,'PassbandRipple',0.2, ...
         'SampleRate',200e3);
fvtool(hpFilt)
```

Output:



Example 7.5: Design a bandpass FIR with the following specifications:

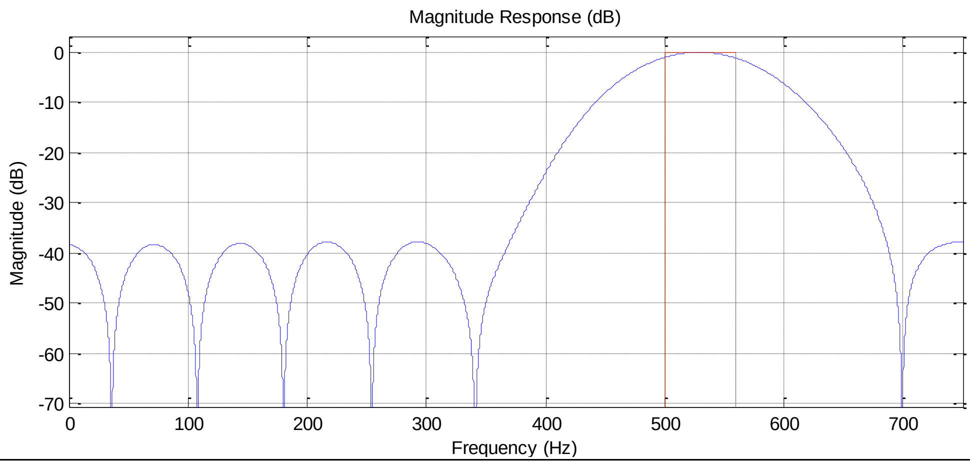
SampleRate: 1500

FilterOrder: 20
CutoffFrequency1: 500
CutoffFrequency2: 560
DesignMethod: 'window'

MATLAB Code:

```
bpFilt = designfilt('bandpassfir','FilterOrder',20, ...
         'CutoffFrequency1',500,'CutoffFrequency2',560, ...
         'SampleRate',1500);
fvtool(bpFilt)
```

Output:



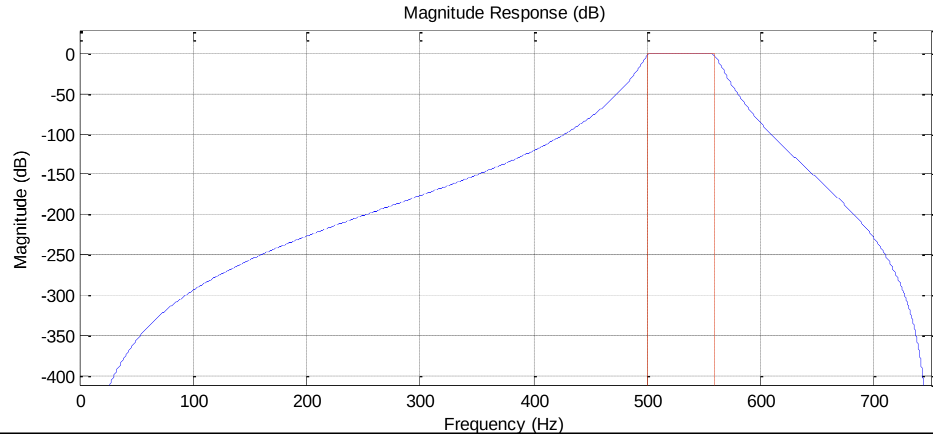
Example 7.6: Design a bandpass IIR with the following specifications:

SampleRate: 1500
FilterOrder: 20
HalfPowerFrequency1: 500
HalfPowerFrequency2: 560
DesignMethod: 'butter'

MATLAB Code:

```
bpFilt = designfilt('bandpassiir','FilterOrder',20, ...
         'HalfPowerFrequency1',500,'HalfPowerFrequency2',560, ...
         'SampleRate',1500)
fvtool(bpFilt)
dataIn = randn([1000 1]); dataOut = filter(bpFilt,dataIn);
```

Output:



Example 7.7: Design a bandstop FIR with the following specifications:

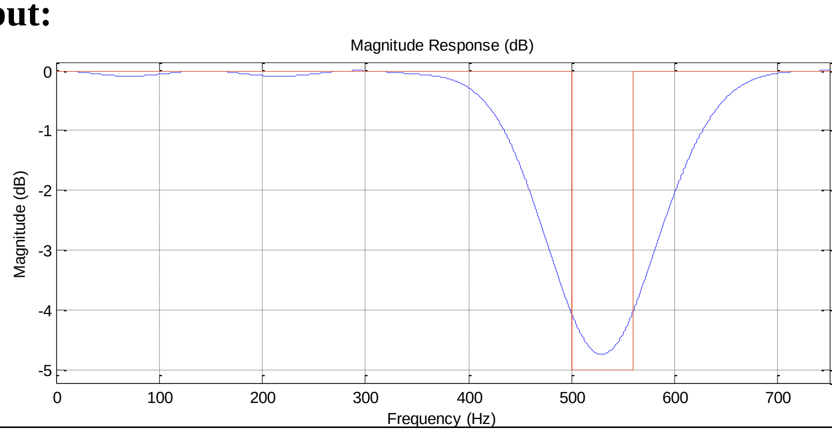
Specifications:

SampleRate: 1500
FilterOrder: 20
CutoffFrequency1: 500
CutoffFrequency2: 560
DesignMethod: 'window'

MATLAB Code:

```
bsFilt = designfilt('bandstopfir','FilterOrder',20, ...
         'CutoffFrequency1',500,'CutoffFrequency2',560, ...
         'SampleRate',1500)
fvtool(bsFilt)
```

Output:



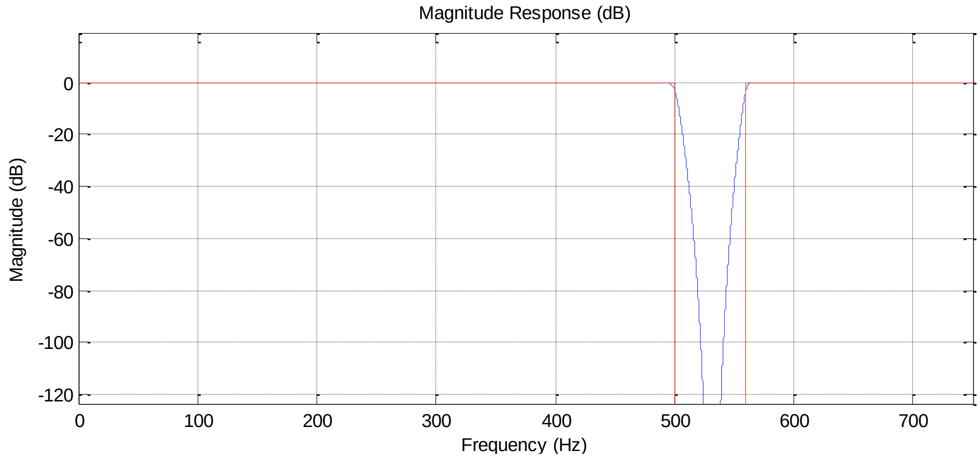
Example 7.8: Design a bandstop IIR with the following specifications:

SampleRate: 1500
FilterOrder: 20
HalfPowerFrequency1: 500
HalfPowerFrequency2: 560
DesignMethod: 'butter'

MATLAB Code:

```
bsFilt = designfilt('bandstopiir','FilterOrder',20, ...
         'HalfPowerFrequency1',500,'HalfPowerFrequency2',560, ...
         'SampleRate',1500)
fvtool(bsFilt)
```

Output:



Lab Session 7.2: Filter Design Using fdatool

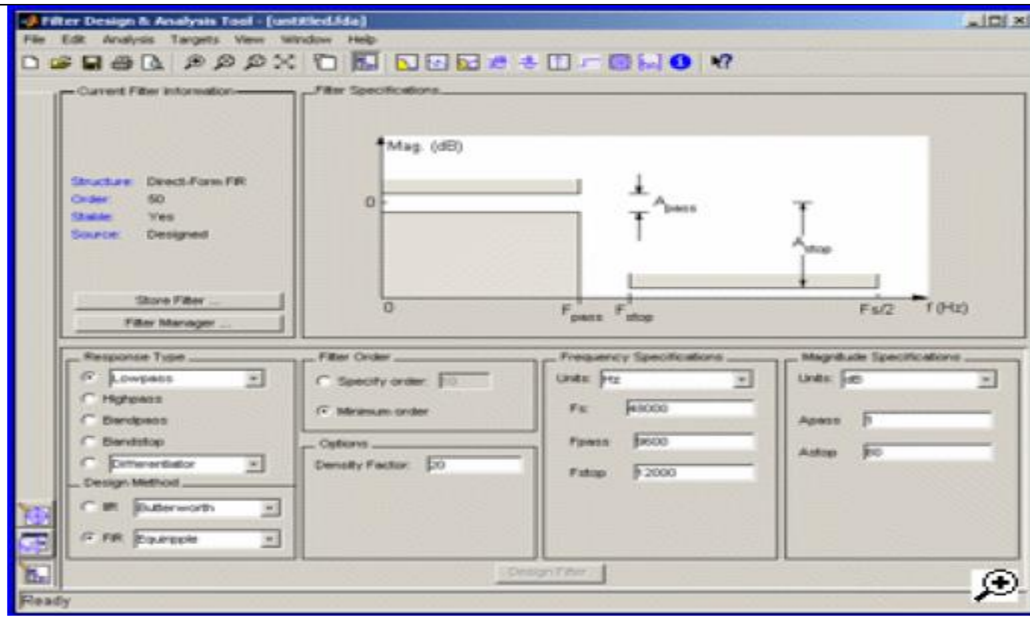
The Filter Design and Analysis Tool (FDATool) is a powerful user interface for designing and analyzing filters quickly. FDATool enables you to design digital FIR or IIR filters by setting filter specifications, by importing filters from your MATLAB workspace, or by adding, moving or deleting poles and zeros. FDATool also provides tools for analyzing filters, such as magnitude and phase response and pole-zero plots.

Introduction to the Filter Design and Analysis Tool (FDATool):

If you type

>>fdatool

in command window, FDAtool will be opened. There you can select FIR or IIR filter, order of filter and cutoff frequency of a filter (either HPF, LPF or BPF). That code will automatically generate .m file for you.



The GUI has three main regions:

- **The Current Filter Information region**
- **The Filter Display region and**
- **The Design panel**

The upper half of the GUI displays information on filter specifications and responses for the current filter. The Current Filter Information region, in the upper left, displays filter properties, namely the filter structure, order, number of sections used and whether the filter is stable or not. It also provides access to the Filter manager for working with multiple filters.

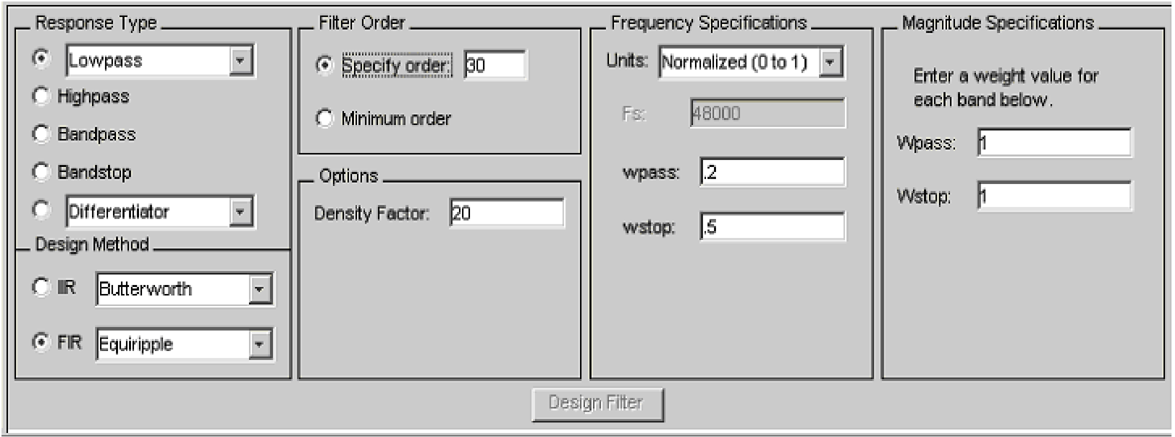
The Filter Display region, in the upper right, displays various filter responses, such as, magnitude response, group delay and filter coefficients.

The lower half of the GUI is the interactive portion of FDATool. The Design Panel, in the lower half is where you define your filter specifications. It controls what is displayed in the other two upper regions. Other panels can be displayed in the lower half by using the sidebar buttons.

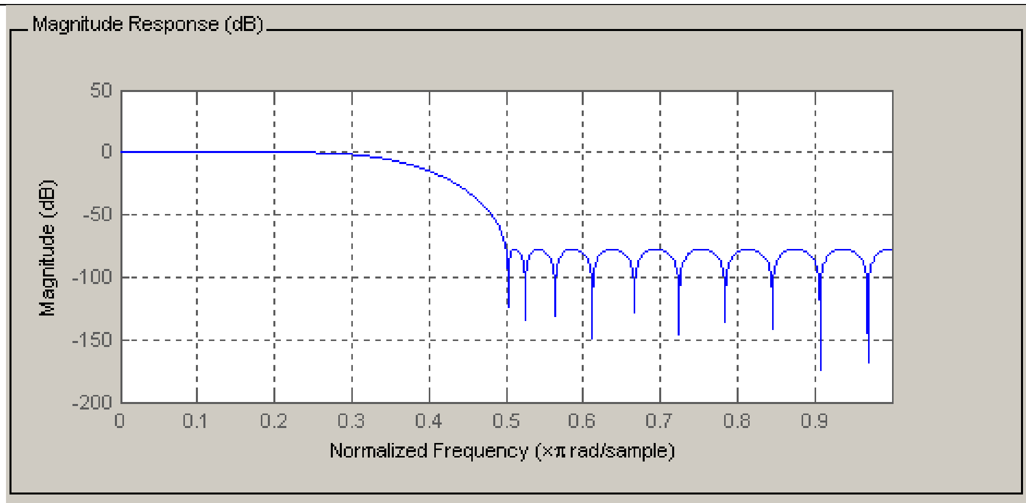
Designing a Filter:

- Low Pass FIR filter
- Filter Order 30
- A passband frequency 0.2 [Normalized (0 to 1)]
- A stopband frequency 0.5 [Normalized (0 to 1)]

To implement this design, we will use the following specifications:

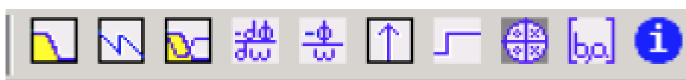


- ❖ Select **Lowpass** from the dropdown menu under **Response Type** and **Equiripple** under **FIR Design Method**.In general, when you change the Response Type or Design Method, the filter parameters and Filter Display region update automatically.
- ❖ Select **Specify order** in the **Filter Order** area and enter **30**.
- ❖ The FIR Equiripple filter has a **Density Factor** option which controls the density of the frequency grid.Increasing the value creates a filter which more closely approximates an ideal equiripple filter, but more time is required as the computation increases. Leave this value at 20.
- ❖ Select **Normalized (0 to 1)** in the Units pull down menu in the **Frequency Specifications** area.
- ❖ Enter **0.2** for **wpass** and **0.5** for **wstop** in the **Frequency Specifications** area.
- ❖ **Wpass** and **Wstop**, in the **Magnitude Specifications** area are positive weights, one per band, used during optimization in the FIR Equiripple filter. Leave these values at 1.
- ❖ After setting the design specifications, click the **Design Filter** button at the bottom of the GUI to design the filter.
- ❖ The magnitude response of the filter is displayed in the Filter Analysis area after the coefficients are computed.



Viewing other Analyses:

Once you have designed the filter, you can view the following filter analyses in the display window by clicking any of the buttons on the toolbar:

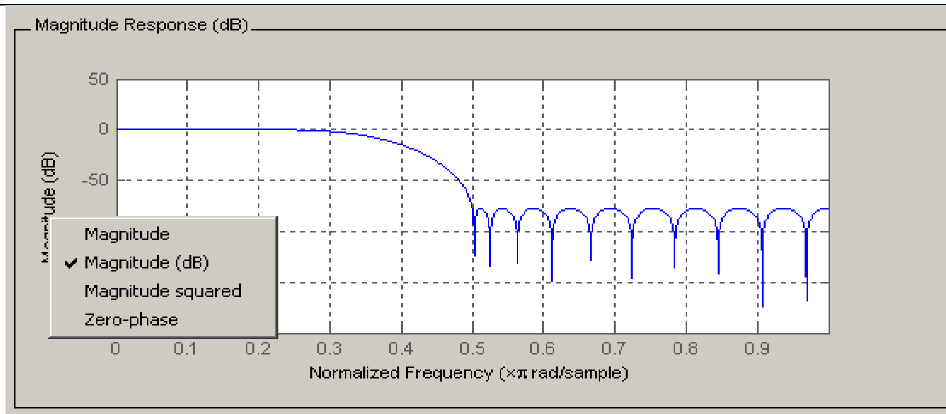


In order from left to right, the buttons are

- Magnitude response
- Phase response
- Magnitude and Phase responses
- Group delay response
- Phase delay response
- Impulse response
- Step response
- Pole-zero plot
- Filter Coefficients
- Filter Information

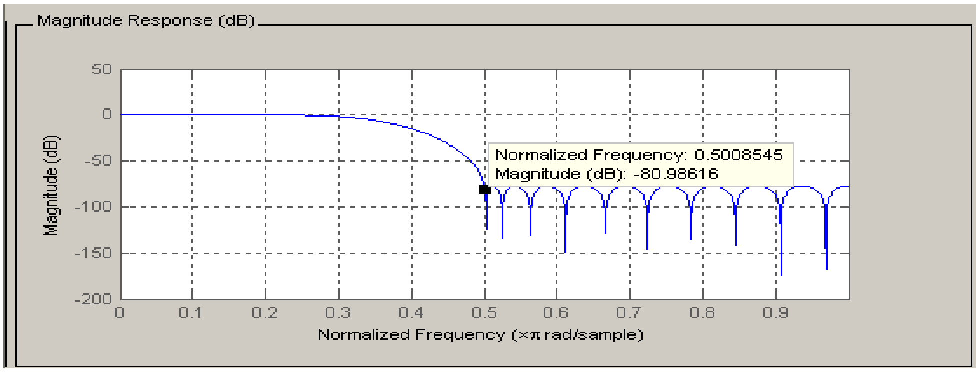
Changing Axes Units:

You can change the x- or y-axis units by right-clicking the mouse on an axis label and selecting the desired units.



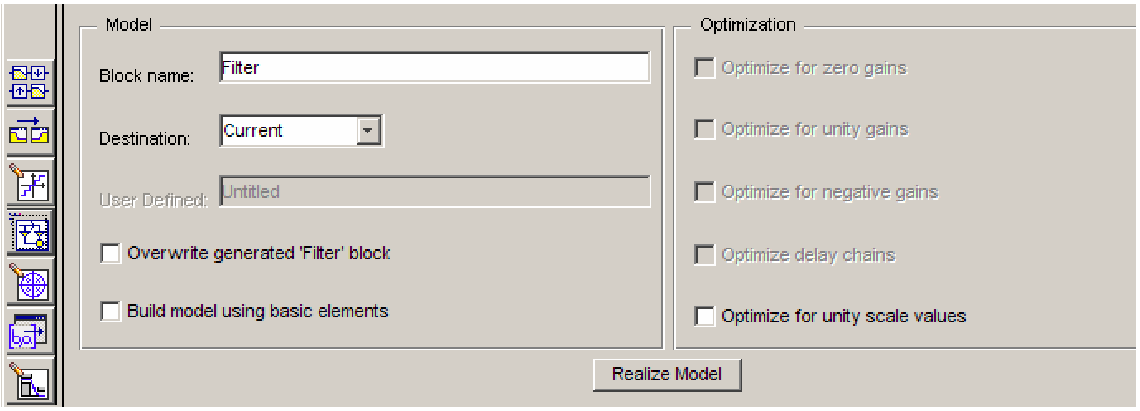
Marking Data Points:

In the Display region, you can click on any point in the plot to add a data marker, which displays the values at that point. Right-clicking on the data marker displays a menu where you can move, delete or adjust the appearance of the data markers.

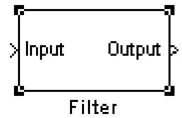


Exporting to a Simulink Model:

1. After designing your filter, click the **Realize Model** sidebar button or select **File** > **Export to Simulink Model**. The Realize Model panel is displayed.



2. Specify the name to use for your block in **Block name**.
3. Select the **Destination** — Current to insert the block into the current (most recently selected) Simulink model or New to open a new model.
4. If you want to overwrite a block previously created from this panel, check **Overwrite generated `Filter' block**.
5. Click the **Realize Model** button to create the filter block. The filter is implemented as a subsystem block using Sum, Gain, and Integer Delay blocks.



Generating an M-File:

FDATool allows you to generate M-code to re-create your filter. This enables you to embed your design into existing code or automate the creation of your filters in a script.
Select **Generate M-file** from the **File** menu and specify the filename in the Generate M-file dialog box.
The following code was generated from the minimum order filter we designed above:

```
%
Fpass = 0.2;              % Passband Frequency
Fstop = 0.5;              % Stopband Frequency
Dpass = 0.057501127785;   % Passband Ripple
Dstop = 0.0001;           % Stopband Attenuation
dens  = 20;               % Density Factor
%
% Calculate the order from the parameters using FIRPMORD.
[N, Fo, Ao, W] = firpmord([Fpass, Fstop], [1 0], [Dpass, Dstop]);
%
% Calculate the coefficients using the FIRPM function.
b  = firpm(N, Fo, Ao, W, {dens});
Hd = dfilt.dffir(b);
% [EOF]
```

Examples of Filter design using fdatool :

Example 7.9

Design an **FIR filter** with the following specifications using Kaiser window :

Passband edge 2 rad/sec
Stopband edge 4 rad/sec
Maximum passband ripple 0.1dB
Minimum stopband attenuation 40 dB
Sampling frequency 20 rad/sec

Show (i) magnitude response (ii) impulse response (iii) Pole zero plot

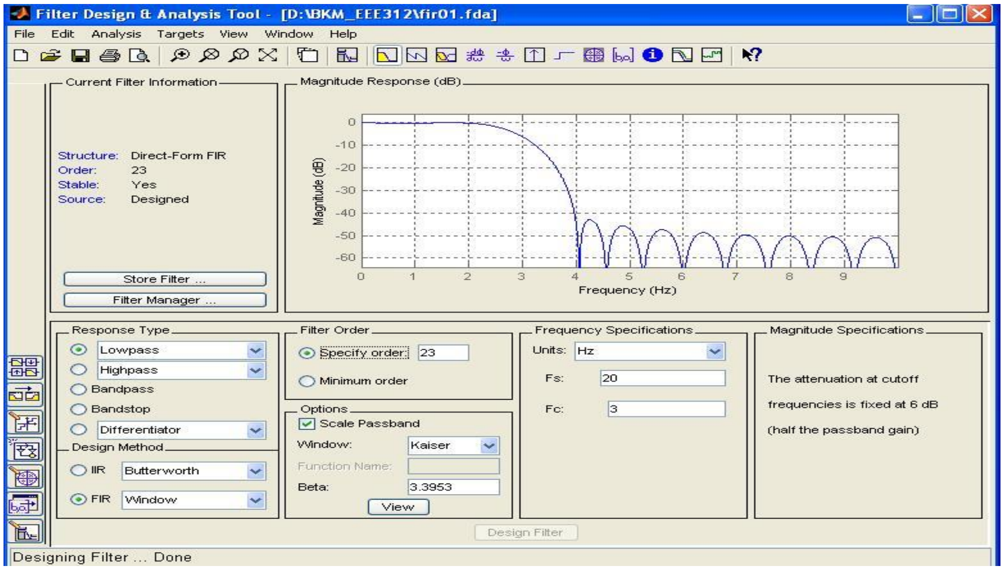
Solution :

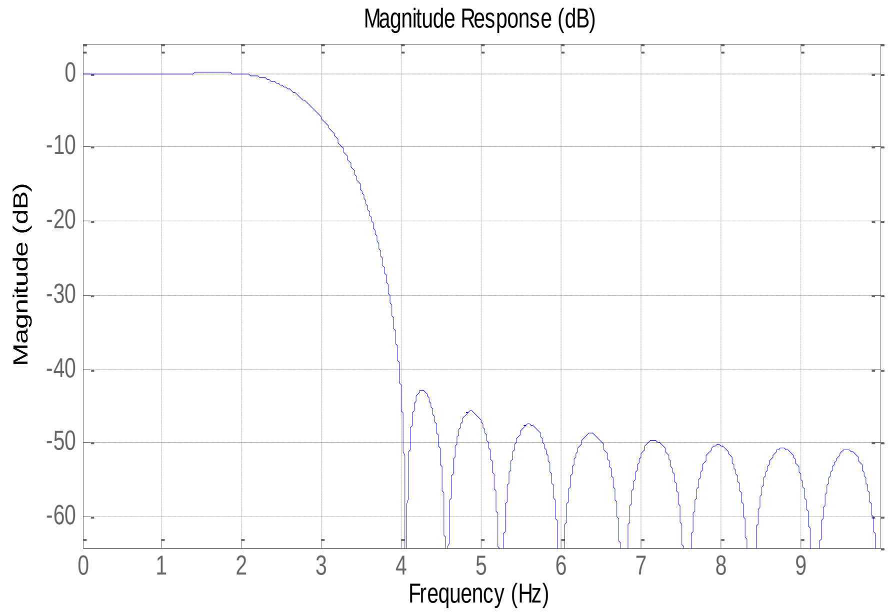
MATLAB code to find the order and Kaiser parameter

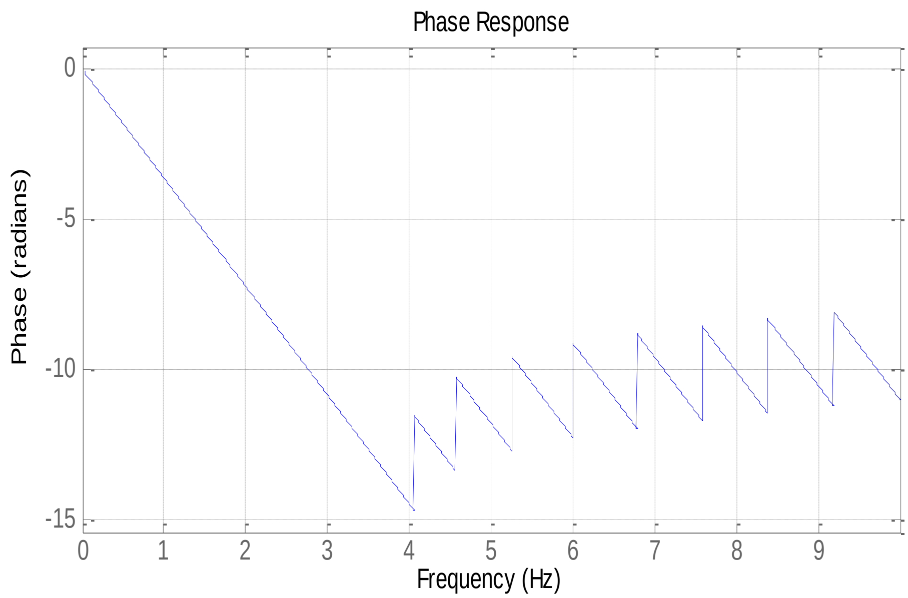
```
fs=20;
f=[2 4];
a=[1 0];
Rp=0.1;
Rs=40;
dp=1-10^(-Rs/20);
ds=10^(-Rs/20);
dev=[dp ds];
[N,Wn,beta,ftype]=kaiserord(f,a,dev,fs)
```

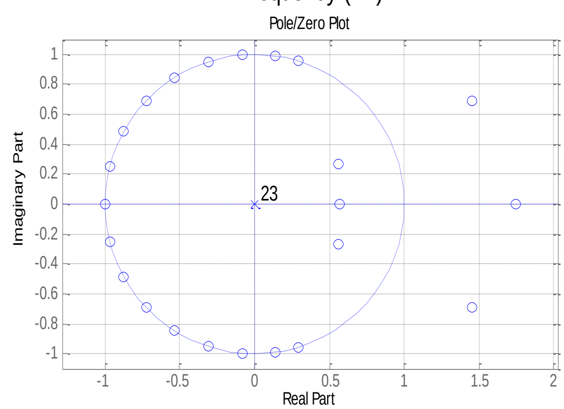
N = 23 and, beta = 3.3953

Now use these data to design the filter.
Insert data as shown in the following figure:









Example 7.10 :

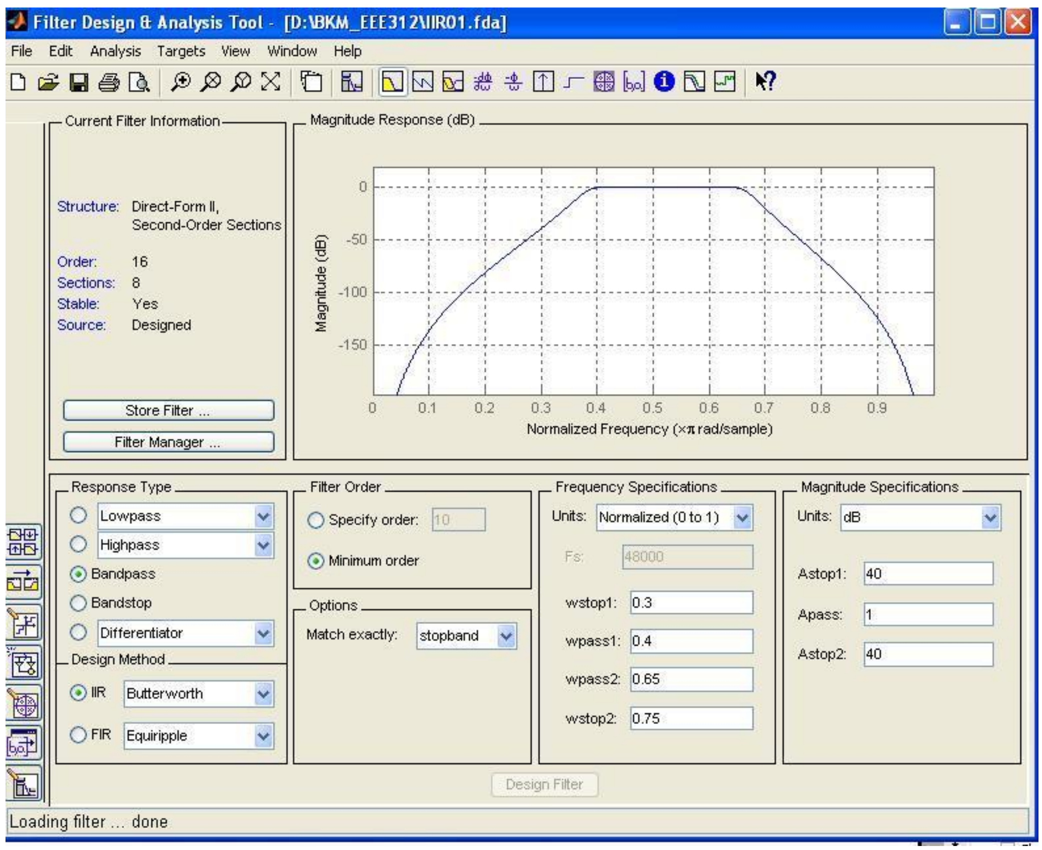
Design an IIR Butterworth bandpass filter with the following specifications :

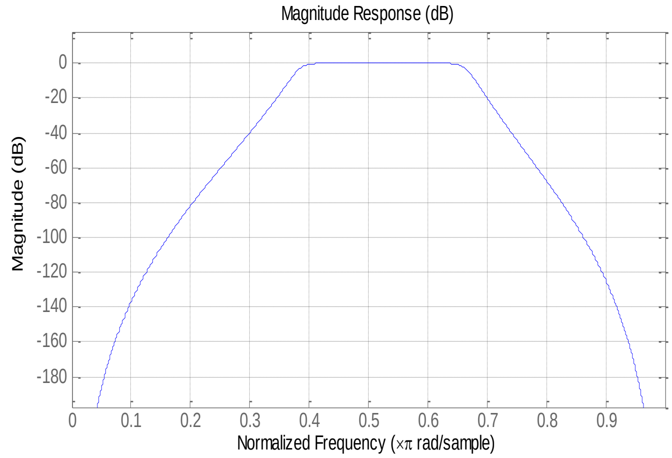
Normalized passband edges are at 0.4 and 0.65
Normalized stopband edges at 0.3 and 0.75
Passband ripple 1 dB
Minimum stopband attenuation 40dB.

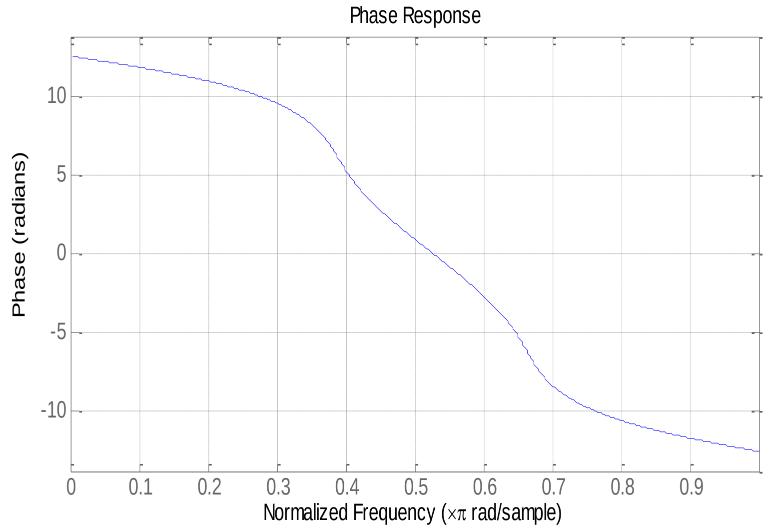
Show (i) Magnitude response (ii) Phase response (iii) Group delay response (iv) Pole zero plot

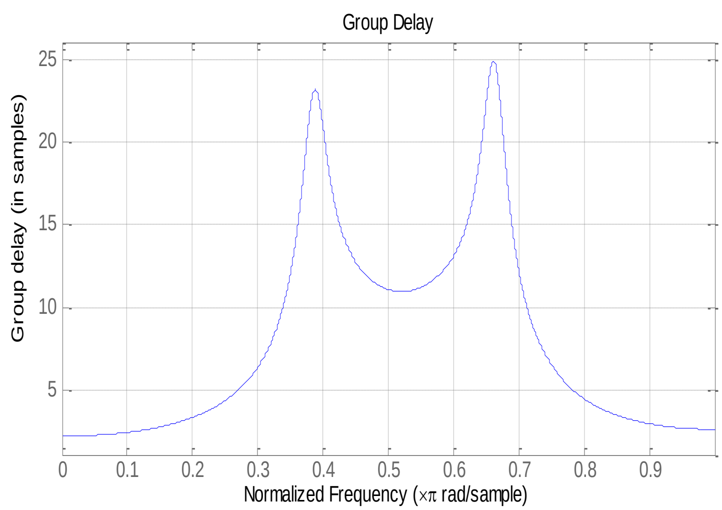
Solution :

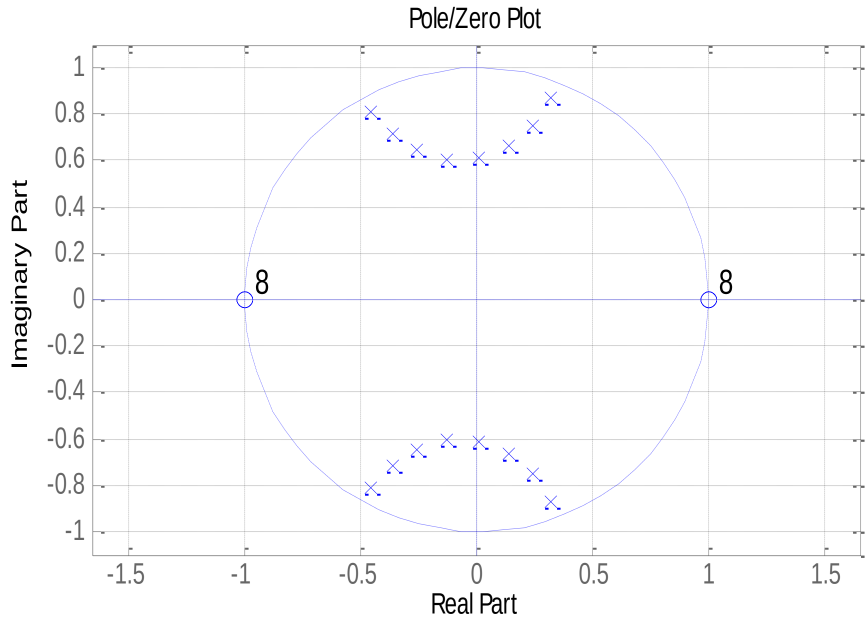
Use the specifications as shown in the figure :




Magnitude Response (dB)
Magnitude (dB)
0
-20
-40
-60
-80
-100
-120
-140
-160
-180
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Normalized Frequency (×π rad/sample)


Phase Response
Phase (radians)
10
5
0
-5
-10
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Normalized Frequency (×π rad/sample)


Group Delay
Group delay (in samples)
25
20
15
10
5
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Normalized Frequency (×π rad/sample)



Lab Session 7.3: In Lab Evaluation

Design a bandstop FIR with the following specifications:

Specifications:

SampleRate: 1000
FilterOrder: 30
CutoffFrequency1: 300
CutoffFrequency2: 500
DesignMethod: 'window'

Home Work:

1. Use designfilt to design FIR and IIR versions of a highpass filter. Specify a normalized stopband frequency of 0.3 and a normalized passband frequency of 0.6. Verify that each filter is of the correct class. Display the frequency responses of the filters.[Hints: Use isfir(fir) and
isfir( ) functions]

2. Use the window method to design a tenth-order lowpass FIR filter with normalized cutoff frequency 0.55. Verify that the filter has linear phase. .[Hints: Use islinphase( )]

3. An ideal low pass filter requires an infinite impulse response. Truncating (or windowing) the impulse response results in the so-called window method of FIR filter design. Consider a simple design of a low pass filter with a cutoff frequency of 0.4*pi radians per sample.